Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

In an apparatus, the electric field was found to oscillate with an amplitude of $24\ \mathrm{V/m}$. The magnitude of the oscillating magnetic field will be

(A) $6 \times 10^{-6}\ \mathrm{T}$

(B) $2 \times 10^{-8}\ \mathrm{T}$

(C) $8 \times 10^{-8}\ \mathrm{T}$

(D) $12 \times 10^{-6}\ \mathrm{T}$

Q.2

A circular ring of radius r is placed in a homogeneous magnetic field perpendicular to the plane of the ring. The field B changes with time according to equation $\mathrm{B} = \mathrm{kt}$, (where, k is a constant and t is the time). The electric field in the ring is:

(A) $\frac{\mathrm{kr}}{4}$

(B) $\frac{\mathrm{kr}}{3}$

(C) $\frac{\mathrm{kr}}{2}$

(D) $\frac{\mathrm{k}}{2\mathrm{r}}$

Q.3

The rms value of the electric field of the light coming from the Sun is $720\ \mathrm{N/C}$. The average total energy density of the electromagnetic wave is

(A) $4.58 \times 10^{-6}\ \mathrm{J/m^3}$

(B) $6.37 \times 10^{-9}\ \mathrm{J/m^3}$

(C) $81.35 \times 10^{-12}\ \mathrm{J/m^3}$

(D) $3.3 \times 10^{-3}\ \mathrm{J/m^3}$

Q.4

For plane electromagnetic waves propagating in the z -direction, which one of the following combination

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gives the correct possible direction
for $\overrightarrow{\mathrm{E}}$ and $\overrightarrow{\mathrm{B}}$ field respectively?

(A) $(2\hat{i}+3\hat{j})$ and $(\hat{i}+2\hat{j})$

(B) $(-2\hat{i}-3\hat{j})$ and $(3\hat{i}-2\hat{j})$

(C) $(3\hat{i}+4\hat{j})$ and $(4\hat{i}-3\hat{j})$

(D) $(\hat{i}+2\hat{j})$ and $(2\hat{i}-\hat{j})$

Q.5

A plane electromagnetic wave propagating in the X-direction has wavelength of $6.0\ \mathrm{mm}$. The electric field is in the Y-direction and its maximum magnitude is $33\mathrm{Vm}^{-1}$. The equation for the electric field as a function of x and t is

(A) $11\sin\pi\left(\mathrm{t}-\frac{\mathrm{x}}{\mathrm{c}}\right)$

(B) $33\sin\left[\pi\times10^{11}\left(\mathrm{t}-\frac{\mathrm{x}}{\mathrm{c}}\right)\right]$

(C) $33\sin\pi\left(\mathrm{t}-\frac{\mathrm{x}}{\mathrm{c}}\right)$

(D) $11\sin\left[\pi\times10^{11}\left(\mathrm{t}-\frac{\mathrm{x}}{\mathrm{c}}\right)\right]$

Q.6

A new system of unit is evolved in which the values of μ_0 and $\in_0$ are 2 and 8 respectively. Then the speed of light in this system will be

(A) 0.25

(B) 0.5

(C) 0.75

(D) 1

Q.7

An electromagnetic wave of frequency 1×10^{14} hertz is propagating along z-axis. The amplitude of electric field is $4\ \mathrm{V/m}$. If $\varepsilon_0=8.8\times10^{-12}\mathrm{C}^2/\mathrm{N}-\mathrm{m}^2$
then average energy density of electric field will be:

(A) $35.2\times10^{-10}\ \mathrm{J/m^3}$

(B) $35.2\times10^{-11}\ \mathrm{J/m^3}$

(C) $35.2\times10^{-12}\ \mathrm{J/m^3}$

(D) $35.2\times10^{-13}\ \mathrm{J/m^3}$

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Q.8

In an electromagnetic wave, the electric and magnetic fields are $100\ \mathrm{V\ m^{-1}}$ and $0.265\ \mathrm{A\ m^{-1}}$. The maximum energy flow is

(A) $26.5\ \mathrm{W/m^2}$

(B) $36.5\ \mathrm{W/m^2}$

(C) $46.7\ \mathrm{W/m^2}$

(D) $765\ \mathrm{W/m^2}$

Q.9

If ε_0 and μ_0 are the electric permittivity and magnetic permeability in vacuum, ε and μ are corresponding quantities in medium, then refractive index of the medium is

(A) $\sqrt{\frac{\varepsilon}{\varepsilon_0}}$

(B) $\sqrt{\frac{\varepsilon_0\mu}{\varepsilon\mu_0}}$

(C) $\sqrt{\frac{\varepsilon_0\mu_0}{\varepsilon\mu}}$

(D) $\sqrt{\frac{\varepsilon\mu}{\varepsilon_0\mu_0}}$

Q.10

A point source of electromagnetic radiation has an average power output of $800\ \mathrm{W}$. The maximum value of electric field at a distance $4.0\ \mathrm{m}$ from the source is

(A) $64.7\ \mathrm{V/m}$

(B) $57.8\ \mathrm{V/m}$

(C) $56.72\ \mathrm{V/m}$

(D) $54.77\ \mathrm{V/m}$

Q.11

In a plane electromagnetic wave propagating in space has an electric field of amplitude $9 \times 10^3\ \mathrm{V/m}$, then the amplitude of the magnetic field is

(A) $2.7 \times 10^{12}\ \mathrm{T}$

(B) $9.0 \times 10^{-3}\ \mathrm{T}$

(C) $3.0 \times 10^{-4}\ \mathrm{T}$

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(D) $3.0 \times 10^{-5}\ \mathrm{T}$

Q.12

If c is the speed of electromagnetic waves in vacuum, its speed in a medium of dielectric constant K and relative permeability μ_{r} is

(A) $\mathrm{v}=\frac{1}{\sqrt{\mu_{\mathrm{r}} \mathrm{K}}}$

(B) $\mathrm{v}=\mathrm{c} \sqrt{\mu_{\mathrm{r}} \mathrm{K}}$

(C) $\mathrm{v}=\frac{\mathrm{c}}{\sqrt{\mu_{\mathrm{r}} \mathrm{K}}}$

(D) $\mathrm{v}=\frac{\mathrm{K}}{\sqrt{\mu_{\mathrm{r}} \mathrm{C}}}$

Q.13

An electromagnetic wave with frequency ω and wavelength λ travels in the $+y$ direction. Its magnetic field is along $+x$ -axis. The vector equation for the associated electric field (of amplitude E_0) is

(A) $\vec{E}=-E_0 \cos \left(\omega t+\frac{2 \pi}{\lambda} y\right) \hat{x}$

(B) $\vec{E}=E_0 \cos \left(\omega t-\frac{2 \pi}{\lambda} y\right) \hat{x}$

(C) $\vec{E}=E_0 \cos \left(\omega t-\frac{2 \pi}{\lambda} y\right) \hat{z}$

(D) $\vec{E}=-E_0 \cos \left(\omega t+\frac{2 \pi}{\lambda} y\right) \hat{z}$

Q.14

The magnetic field between the plates of radius $12\ \mathrm{cm}$ separated by distance of $4\ \mathrm{mm}$ of a parallel plate capacitor of capacitance $100 \mathrm{pF}$ along the axis of plates having conduction current of 0.15 A is

(A) zero

(B) $1.5\ \mathrm{T}$

(C) $15\ \mathrm{T}$

(D) $0.15\ \mathrm{T}$

Q.15

Consider an electromagnetic wave propagating in vacuum. Choose the correct statement:

(A) For an electromagnetic wave propagating in $+y$ direction the electric field is $\overrightarrow{\mathrm{E}}=\frac{1}{\sqrt{2}} \mathrm{E}_{\mathrm{yz}}(\mathrm{x}, \mathrm{t}) \hat{\mathrm{z}}$ and the magnetic field is $\overrightarrow{\mathrm{B}}=\frac{1}{\sqrt{2}}\ \mathrm{B}_{\mathrm{Z}}(\mathrm{x}, \mathrm{t}) \hat{\mathrm{y}}$

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(B) For an electromagnetic wave propagating in $+y$ direction the electric field is $\overrightarrow{\mathrm{E}}=\frac{1}{\sqrt{2}}\mathrm{E}_{\mathrm{yz}}(\mathrm{x}, \mathrm{t})\hat{\mathrm{y}}$ and the magnetic field is $\overrightarrow{\mathrm{B}}=\frac{1}{\sqrt{2}}\ \mathrm{B}_{\mathrm{yz}}(\mathrm{x}, \mathrm{t})\hat{\mathrm{z}}$

(C) For an electromagnetic wave propagating in $+\mathrm{x}$ direction the electric field is $\overrightarrow{\mathrm{E}}=\frac{1}{\sqrt{2}}\mathrm{E}_{\mathrm{yz}}(\mathrm{y}, \mathrm{z}, \mathrm{t})(\hat{\mathrm{y}}+\hat{\mathrm{z}})$ and the magnetic field is $\overrightarrow{\mathrm{B}}=\frac{1}{\sqrt{2}}\ \mathrm{B}_{\mathrm{yz}}(\mathrm{y}, \mathrm{z}, \mathrm{t})(\hat{\mathrm{y}}+\hat{\mathrm{z}})$

(D) For an electromagnetic wave propagating in $+x$ direction the electric field is $\overrightarrow{\mathrm{E}}=\frac{1}{\sqrt{2}}\mathrm{E}_{\mathrm{yz}}(\mathrm{x}, \mathrm{t})(\hat{\mathrm{y}}-\hat{\mathrm{z}})$ and the magnetic field is $\overrightarrow{\mathrm{B}}=\frac{1}{\sqrt{2}}\ \mathrm{B}_{\mathrm{yz}}(\mathrm{x}, \mathrm{t})(\hat{\mathrm{y}}+\hat{\mathrm{z}})$

Q.16

An electromagnetic wave of frequency $v=3.0$ MHz passes from vacuum into a dielectric medium with permittivity $\in=4.0$. Then

(A) wavelength is halved and frequency remains unchanged

(B) wavelength is doubled and frequency becomes half

(C) wavelength is doubled and the frequencyremains unchanged

(D) wavelength and frequency both remain unchanged.

Q.17

A plane electromagnetic wave is incident on a plane surface of area A, normally and is perfectly reflected. If energy E strikes the surface in time t then force exerted on the surface is $(\mathrm{c}=$ speed of light $)$

(A) $\frac{2\mathrm{E}}{\mathrm{At}}$

(B) $\frac{\mathrm{E}}{2\mathrm{c}}$

(C) $\frac{2\mathrm{E}}{\mathrm{ct}}$

(D) zero

Q.18

If a source is transmitting electromagnetic wave of frequency $8.2 \times 10^{6}\ \mathrm{Hz}$, then wavelength of the electromagnetic waves transmitted from the source will be

(A) $36.6\ \mathrm{m}$

(B) $40.5\ \mathrm{m}$

(C) $42.3\ \mathrm{m}$

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(D) $50.9\ \mathrm{m}$

Q.19

Which of the following waves have the maximum wavelength

(A) X-rays

(B) I.R. rays

(C) UV rays

(D) Radio waves

Q.20

The frequency of the AM range of radio waves is _____

(A) less than 30 MHz

(B) more than 30 MHz

(C) less than 20000 Hz

(D) more than 20000 Hz

Answers & Solutions

Q.1 Answer:

$8 \times 10^{-8}\ \mathrm{T}$

Solution:

$$\frac{E}{B} = C \Rightarrow B = \frac{E}{C} = \frac{24}{3\times10^8} = 8 \times 10^{-8}\ \mathrm{T}$$

Q.2 Answer:

$\frac{\mathrm{kr}}{2}$

Solution:

$$\oint \vec{E} \cdot d\vec{\ell} = -\frac{d\phi_B}{dt}$$
$$\text{or } \mathrm{E}(2\pi \mathrm{r}) = \frac{d}{dt}\left(kt \times \pi r^2\right) \quad \therefore \quad \mathrm{E} = \frac{\mathrm{kr}}{2}$$

Q.3 Answer:

$4.58 \times 10^{-6}\ \mathrm{J/m^3}$

Solution:

$$\mathrm{E_{rms}} = 720$$

$$\textbf{The average total energy density} = \frac{1}{2}\epsilon_0 E_0^2 = \frac{1}{2}\epsilon_0 \left[\sqrt{2}E_{\mathrm{rms}}\right]^2 = \epsilon_0 E_{\mathrm{rms}}^2$$

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$= 8.85 \times 10^{-12} \times (720)^2 = 4.58 \times 10^{-6}\ \mathrm{J/m^3}$

Q.4 Answer:

$(-2\hat{i} - 3\hat{j})$ and $(3\hat{i} - 2\hat{j})$

Solution:

As we know, $\vec{E} \cdot \vec{B} = 0 \quad \because [\vec{E} \perp \vec{B}]$
and $\vec{E} \times \vec{B}$ **should be along** Z **direction As** $(-2\hat{i} - 3\hat{j}) \times (3\hat{i} - 2\hat{j}) = 5\hat{k}$
Hence option (b) is the correct answer.

Q.5 Answer:

$33 \sin\left[\pi \times 10^{11}\left(\mathrm{t} - \frac{\mathrm{x}}{\mathrm{c}}\right)\right]$

Solution:

$\omega = 2\pi v = \frac{2\pi c}{\lambda} = \frac{2\pi \times 3 \times 10^8}{6 \times 10^{-3}}$
$= \pi \times 10^{11} \mathrm{rad/sec}$
The equation for the electric field, along y-axis in the electromagnetic wave is $\mathrm{Ey} = \mathrm{E_0} \sin\omega\left(t - \frac{x}{c}\right) = 33 \sin\left[\pi \times 10^{11}\left(t - \frac{x}{c}\right)\right]$

Q.6 Answer:

0.25

Solution:

The speed of light $C = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = \frac{1}{\sqrt{2 \times 8}} = \frac{1}{4} = 0.25$

Q.7 Answer:

$35.2 \times 10^{-12}\ \mathrm{J/m^3}$

Solution:

Given: Amplitude of electric field, $E_0 = 4\mathrm{v/m}$
Absolute permitivity, $\varepsilon_0 = 8.8 \times 10^{-12}\mathrm{c^2/N - m^2}$
Average energy density $u_E =$ **? Applying formula, Average energy density** $u_E = \frac{1}{4}\varepsilon_0 E^2$
$\Rightarrow \quad u_E = \frac{1}{4} \times 8.8 \times 10^{-12} \times (4)^2 = 35.2 \times 10^{-12}\ \mathrm{J/m^3}$

Q.8 Answer:

$26.5\ \mathrm{W/m^2}$

Solution:

Here, amplitude of electric field, $\mathrm{E_0} = 100\ \mathrm{V/\ m}$**; amplitude of magnetic field,** $\mathrm{H_0} = 0.265\ \mathrm{A/m}$**. We know that the maximum rate of energy flow,** $\mathrm{S} = \mathrm{E_0} \times \mathrm{H_0} = 100 \times 0.265 =$

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$26.5\ \mathrm{W/m^2}$

Q.9 Answer:

$\sqrt{\frac{\varepsilon\mu}{\varepsilon_0\mu_0}}$

Solution:

No solution available

Q.10 Answer:

$54.77\ \mathrm{V/m}$

Solution:

Intensity of EM wave is given by

$I = \frac{P}{4\pi R^2} = v_{av} \cdot c = \frac{1}{2}\varepsilon_0 E_0^2 \times c$

$\Rightarrow E_0 = \sqrt{\frac{P}{2\pi R^2 \varepsilon_0 c}}$

$= \sqrt{\frac{800}{2\times 3.14\times (4)^2 \times 8.85\times 10^{-12}\times 3\times 10^8}} = 54.77 \frac{V}{m}$

Q.11 Answer:

$3.0 \times 10^{-5}\ \mathrm{T}$

Solution:

$B_0 = \frac{E_0}{c} = \frac{9\times 10^3}{3\times 10^8} = 3 \times 10^{-5} T.$

Q.12 Answer:

$\mathrm{v} = \frac{\mathrm{c}}{\sqrt{\mu_\mathrm{r}\mathrm{K}}}$

Solution:

Speed of light of vacuum $c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}$ **and in another medium** $v = \frac{1}{\sqrt{\mu\varepsilon}}$

$\therefore \quad \frac{c}{v} = \sqrt{\frac{\mu\varepsilon}{\mu_0\varepsilon_0}} = \sqrt{\mu_r K} \Rightarrow v = \frac{c}{\sqrt{\mu_r K}}.$

Q.13 Answer:

$\vec{E} = E_0 \cos\left(\omega t - \frac{2\pi}{\lambda} y\right) \hat{z}$

Solution:

In an electromagnetic wave electric tield and magnetic field are perpendicular to the direction of propagation of wave. The vector equation for the electric field is

$\vec{E} = E_0 \cos\left(\omega t - \frac{2\pi}{\lambda} y\right) \hat{z}$

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Q.14 Answer:

zero

Solution:

As $\mathrm{B} \propto \mathrm{r}$, since the point is on the axis where $\mathrm{r}=0$, So $\mathrm{B}=0$

Q.15 Answer:

For an electromagnetic wave propagating in $+x$ direction the electric field is $\overrightarrow{\mathrm{E}}=\frac{1}{\sqrt{2}} \mathrm{E}_{\mathrm{yz}}(\mathrm{x}, \mathrm{t})(\hat{\mathrm{y}}-\hat{\mathrm{z}})$ and the magnetic

field is $\overrightarrow{\mathrm{B}}=\frac{1}{\sqrt{2}} \mathrm{B}_{\mathrm{yz}}(\mathrm{x}, \mathrm{t})(\hat{\mathrm{y}}+\hat{\mathrm{z}})$

Solution:

Wave in X-direction means E and B should be function of x and t.

$\hat{y}-\hat{z} \perp \hat{y}+\hat{z}$

Q.16 Answer:

wavelength is halved and frequency remains unchanged

Solution:

Frequency remains constant during refraction

$v_{\text{med}}=\frac{1}{\sqrt{\mu_0 \in_0 \times 4}}=\frac{c}{2}$

$\frac{\lambda_{\text{med}}}{\lambda_{\text{air}}}=\frac{v_{\text{med}}}{v_{\text{air}}}=\frac{c/2}{c}=\frac{1}{2}$

$\therefore$ wavelength is halved and frequency remains unchanged.

Q.17 Answer:

$\frac{2\mathrm{E}}{\mathrm{ct}}$

Solution:

Incident momentum, $\mathrm{p}=\frac{E}{c}$

For perfectly reflecting surface with normal incidence

$\Delta \mathrm{p}=2 \mathrm{p}=\frac{2E}{c}; \quad F=\frac{\Delta p}{\Delta t}=\frac{2E}{ct}$

Q.18 Answer:

$36.6 \mathrm{~m}$

Solution:

$\lambda=\frac{c}{v}=\frac{3 \times 10^8}{8.2 \times 10^6}=36.5 \mathrm{~m}$

Q.19 Answer:

Radio waves

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Solution:

$\lambda_{\text{Radiowave}} > \lambda_{\text{IR rays}} > \lambda_{\text{UV rays}} > \lambda_{\text{x-rays}}$

Q.20 Answer:

less than 30 MHz

Solution:

AM radio waves are used to relay commercial radio signals within the 540 **to** 1600 kHz **frequency range.**

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